Lecture 9: Limits and Continuity in Several Variables

Background & Motivation

Before we can differentiate multivariate functions, we need limits. The key new difficulty: in one variable, you approach a point from the left or right. In two variables, you can approach from infinitely many directions. This makes proving limits exist much harder—and gives us a powerful tool (path analysis) for showing they don't.

1. Multivariable Limits

Definition 1: Limit of $f(x,y)$ as $(x,y) \to (a,b)$

We write $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if $f(x,y)$ can be made arbitrarily close to L for all (x,y) sufficiently close to (a,b).

Key difference from single-variable:

- In 1D – only ____________ (left and right)
- In 2D – must approach from ________________________
- If *any two paths* give different limits $\implies$ limit **DNE**

Properties of Multivariable Limits

If $\lim_{(x,y)\to(a,b)} f = L$ and $\lim_{(x,y)\to(a,b)} g = M$ exist:

- $\lim(f \pm g) = L \pm M$
- $\lim(cf) = cL$
- $\lim(fg) = LM$
- $\lim(f/g) = L/M$ if $M \neq 0$

Polynomials and rational functions (where defined) – use **direct substitution**.

2. Techniques for Evaluating Limits

Technique 1: Direct Substitution

– Always try first; works when f is continuous at (a,b)

Technique 2: Algebraic Simplification / Rationalization

– Factor, multiply by conjugate, or simplify before substituting

Example 1: Rationalization

Evaluate $\displaystyle\lim_{(x,y)\to(1,1)} \frac{\sqrt{x}-\sqrt{y}}{x-y}$.

Technique 3: Polar / Spherical Coordinates

– For limits at the origin: $x = r\cos\theta$, $y = r\sin\theta$, $(x, y) \to (0, 0) \iff r \to 0^+$

– If the result is independent of θ and tends to L as $r \to 0$, the limit is L

Example 2: Polar Coordinates

Evaluate $\displaystyle\lim_{(x,y,z)\to(0,0,0)} \frac{\sin\left(x^2+y^2+z^2\right)}{x^2+y^2+z^2}$.

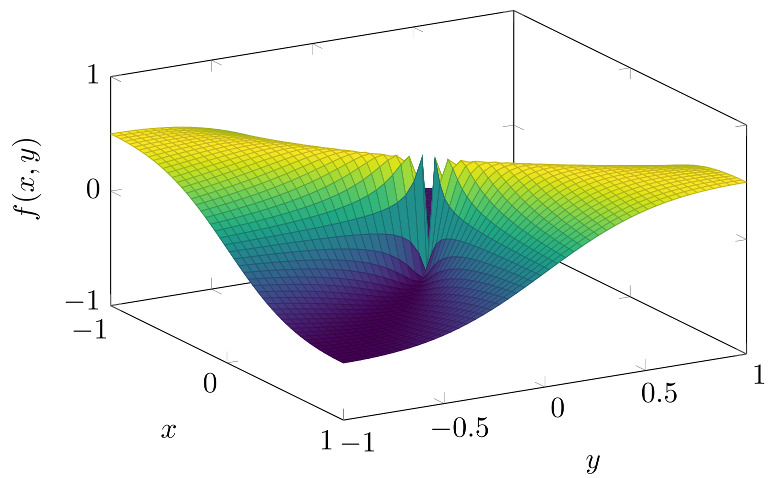
Technique 4: Path Analysis (for showing DNE)

– If two different paths give different limiting values $\implies$ limit DNE

Common Paths to Try

- Along axes: $y = 0$, then $x = 0$
- Along lines: $y = mx$ for various m
- Along curves: $y = x^2$, $y = x^3$, etc.

If all linear paths give the same value – try a curved path before concluding the limit exists!



Graph of $z = \dfrac{xy}{x^2+y^2}$ near the origin. The surface takes different limiting values along different paths to $(0,0)$: along $y=0$ or $x=0$ it equals 0, but along $y=x$ it equals $1/2$. Hence the limit does not exist.

Example 3: Limit DNE by Path Analysis

Show $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

Example 4: Tricky: All Lines Give 0, But Limit DNE

Does $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4}$ exist?

3. Continuity

Definition 2: Continuity of $f(x, y)$

f is **continuous at** (a, b) if all three hold:

1. $f(a, b)$ is defined
2. $\lim_{(x,y)\to(a,b)} f(x, y)$ exists
3. $\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$

Practice Problems

Problem 1. Evaluate $\lim_{(x,y)\to(2,3)} (x^2y + xy^2 - 1)$.

Problem 2. Evaluate $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2 + y^2}$.

Problem 3. Show $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

Problem 4. Evaluate $\lim_{(x,y)\to(1,1)} \frac{x^2 - y^2}{x - y}$.

Problem 5. Show $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2y^2}{x^2y^2+(x-y)^2}$ does not exist.

Problem 6. Prove using ϵ-δ that $\lim\limits_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^2+y^2} = 0$.

Hint: Use $|y| \leq \sqrt{x^2+y^2}$.